

BUSINESS ANALYSIS REPORT

Maven Toys

Sales, Profitability & Inventory Analysis — Mexico Retail Network

Prepared for: Maven Toys Executive & Merchandising Team

Scope: 50 stores · 29 cities · 35 SKUs across 5 categories

Period Covered: January 1, 2022 – September 30, 2023

Report Date: July 2026

Executive Summary

This report analyzes 21 months of point-of-sale transactions (January 2022 – September 2023) across Maven Toys' 50-store retail network in Mexico, covering 829,262 individual sales records, 1,090,565 units sold, and \$14.44M in gross revenue. The analysis was built to answer the four business questions Maven Analytics identified as the core objectives of this guided project: category profitability by location, seasonal sales patterns, stockout-driven revenue loss, and inventory carrying cost.

Across the full period, Maven Toys generated \$4.01M in gross profit at a 27.8% blended margin. Toys is the highest-revenue category (\$5.09M) but Electronics is the most profitable on a margin basis (44.6% margin vs. 21.2% for Toys), despite carrying only 3 of the store's 35 SKUs. Downtown stores are the clear profit engine of the network, outperforming Commercial, Residential, and Airport locations across every product category without exception.

Sales show a consistent, repeatable seasonal pattern: a strong December holiday peak in both years (Dec 2022: \$877K, the single highest month in the dataset), a spring rebound each March–April, and a mid-summer trough each July–August. Weekends — particularly Friday and Saturday — generate materially more revenue than weekdays, together accounting for 39% of total revenue despite being 2 of 7 days.

On inventory, 4.8% of all store-SKU combinations (77 of 1,593) are currently completely out of stock, with Toys the most exposed category at a 7.4% stockout rate — more than four times the rate seen in Games (1.8%). The network carries roughly 17 days of supply on hand at current sell-through rates and \$300.2K in inventory value at cost, but that supply is unevenly distributed: some SKUs hold under 7 days of cover while others sit at nearly 58 days, pointing to a mismatch between purchasing cadence and actual demand at the SKU level.

Methodology & Data Sources

This analysis uses the Maven Toys dataset (Maven Analytics, public domain), consisting of five relational tables: Sales (829,262 transactions), Products (35 SKUs across 5 categories), Stores (50 locations across 29 Mexican cities), Inventory (1,593 store-SKU stock records), and a Calendar dimension table. Sales were joined to Products and Stores to derive per-transaction Revenue (Units × Product Price) and Profit (Units × (Product Price – Product Cost)).

Findings reconcile against an in-progress Power BI dashboard build for this project: the September 2023 monthly slice independently shows 41,830 orders, \$658.19K in revenue, and \$180.45K in profit — matching this report's month-level detail exactly, confirming the data model and DAX measures are calculating correctly before the full dashboard is finalized.

Table	Records	Key Fields	Role
Sales	829,262	Date, Store_ID, Product_ID, Units	Fact table
Products	35	Category, Cost, Price	Dimension
Stores	50	City, Location Type, Open Date	Dimension
Inventory	1,593	Stock_On_Hand by Store × Product	Snapshot fact

Sales Performance Overview

Headline KPIs across the full 21-month analysis window.

TOTAL REVENUE	GROSS PROFIT	TOTAL ORDERS	STORE NETWORK
\$14.44M	\$4.01M	829,262	50 stores
Jan 2022 – Sep 2023	27.8% margin	1.09M units sold	29 cities, 4 location types

Question 1 — Which categories drive the biggest profits, and does this hold across locations?

Toys is the largest category by revenue (\$5.09M, 35% of total) but ranks last by margin at 21.2%. Electronics, despite generating the least revenue of any category (\$2.25M) from just 3 SKUs, delivers the highest profit margin at 44.6% and is effectively tied with Toys for total profit dollars (\$1.00M vs. \$1.08M). Games and Art & Crafts sit in the middle at 30.3% and 27.8% margin respectively, while Sports & Outdoors is the weakest performer on both revenue and margin.

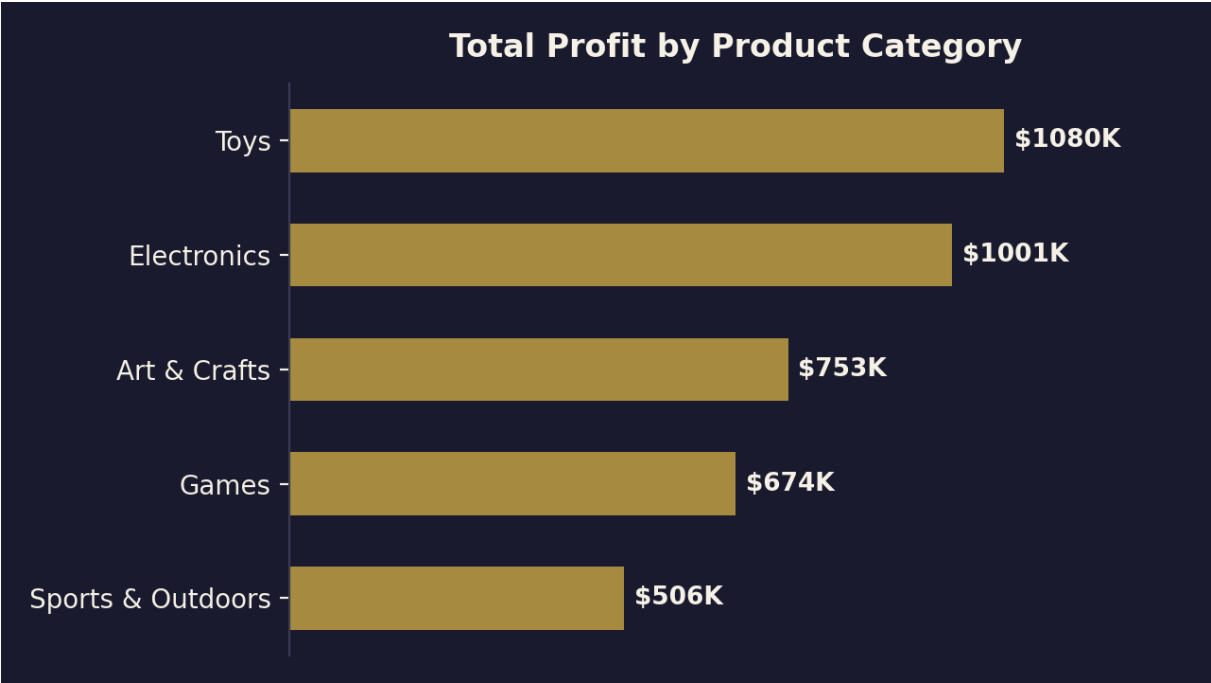


Figure 1. Total profit by product category, full period.

Breaking this down by store location type shows the pattern is highly consistent: Downtown stores are the top profit generator in every single category, followed by Commercial, then a close race between Residential and Airport depending on category. This consistency is the key finding — there is no category that over- or under-performs specifically in one location type relative to the others; Downtown's advantage appears to be a scale effect (more foot traffic, higher basket volume) rather than a category-specific preference.

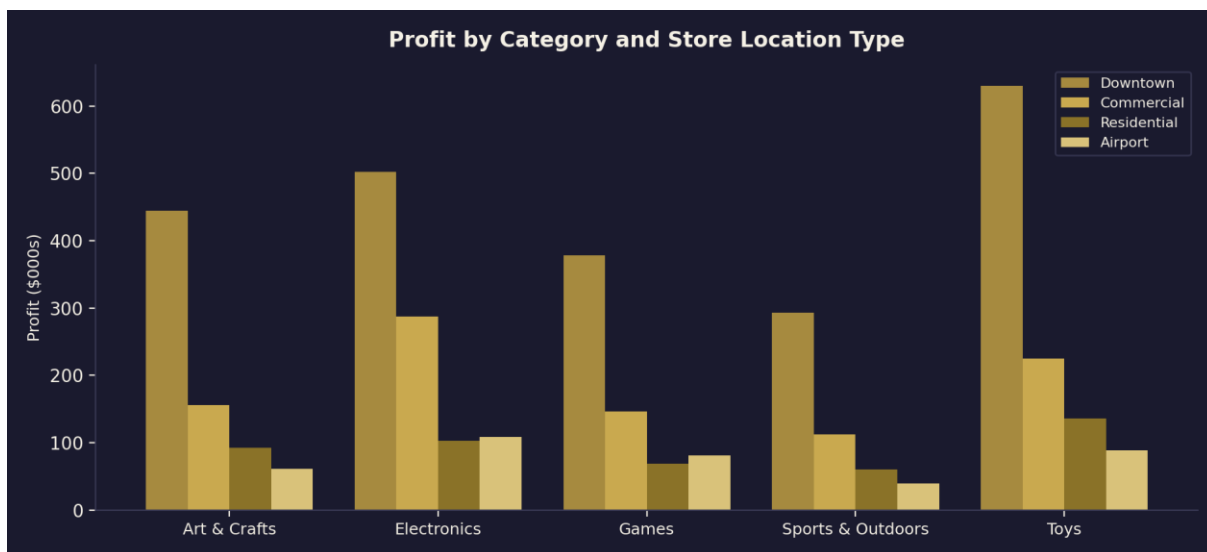


Figure 2. Profit by category, broken out by store location type.

- Electronics is the clearest margin opportunity: 44.6% margin on just 3 SKUs suggests room to expand the assortment without diluting profitability.
- Downtown stores generate more profit than Commercial, Residential, and Airport combined in most categories — a strong signal for where to prioritize new store openings or renovation capital.
- Sports & Outdoors is the one category worth a pricing or cost review — it trails every other category on margin despite reasonable revenue scale.

Question 2 — Are there seasonal trends or patterns in the sales data?

Yes — a clear and repeatable seasonal cycle is visible across both full years in the dataset. December is the strongest month every year (Dec 2022: \$877.2K, +33% over the trailing 3-month average), consistent with holiday gift-buying. A secondary spring peak occurs each March, and a mid-summer dip appears reliably in July–August of both years, likely tied to the school calendar and reduced discretionary spend before the September back-to-school period.



Figure 3. Monthly revenue trend, January 2022 – September 2023.

At a weekly level, Friday and Saturday are consistently the two highest-revenue days, together contributing 39% of total revenue across the full period — nearly double what a proportional 2-of-7-day split would predict. Monday is the softest day of the week.

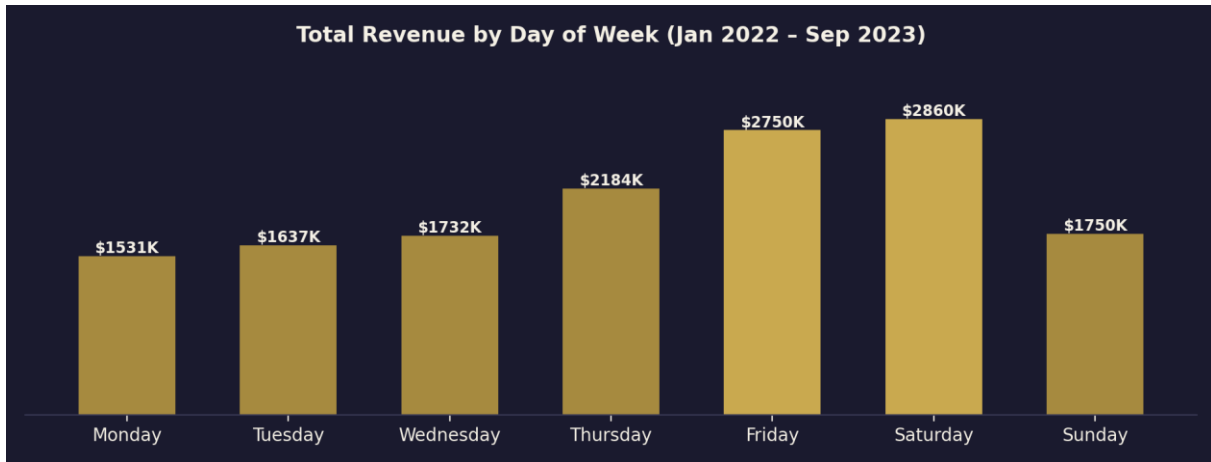


Figure 4. Total revenue by day of week, full period.

- December promotional and staffing plans should be locked in early — it is the single most important month of the year for revenue capture.
- Weekend staffing and inventory replenishment should be weighted toward Friday/Saturday, where nearly 2 in 5 revenue dollars are earned.
- The July–August trough is a natural window for clearance promotions or inventory rebalancing ahead of the September–December ramp.

Question 3 — Are sales being lost to out-of-stock products at certain locations?

Yes. As of the current inventory snapshot, 77 of 1,593 store-SKU combinations (4.8%) are completely out of stock — meaning those products are generating zero sales at that location until restocked. The risk is not evenly distributed: Toys has the highest stockout rate at 7.4%, more than four times the rate in Games (1.8%) and nearly three times the rate in Electronics (2.7%).

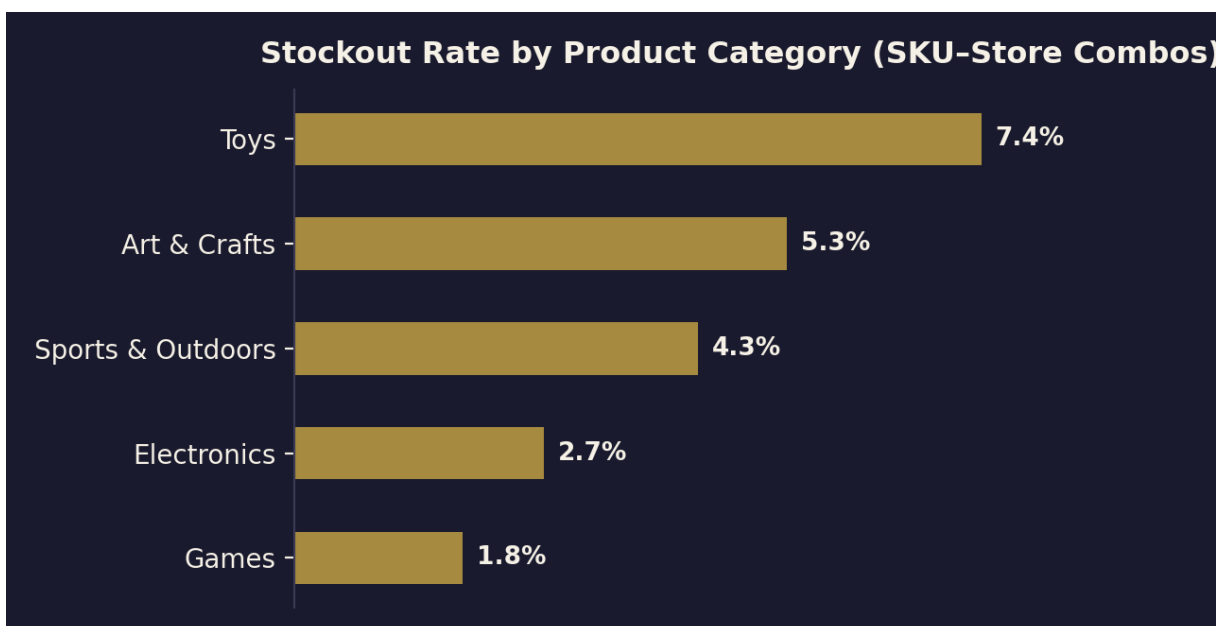


Figure 5. Stockout rate by product category, current inventory snapshot.

This is a meaningful finding given Toys is simultaneously the highest-revenue category in the network — the category most exposed to stockouts is also the category with the most revenue at stake per lost sale-day. Because Toys also carries the thinnest margin of any category (21.2%), each stockout is a doubly costly event: high revenue exposure paired with the lowest per-unit profit cushion.

- Toys and Art & Crafts should be the priority for a minimum-stock-level or safety-stock policy in Power BI inventory alerting.
- A stockout-rate KPI card, filterable by category and store, would let regional managers spot and react to gaps in near real time.
- Given Electronics' high margin and low stockout rate, its replenishment process is a reasonable internal benchmark for the other categories.

Question 4 — How much money is tied up in inventory, and how long will it last?

The network currently holds 29,742 units in stock across all stores and SKUs, valued at \$300,209.58 at cost. At the current network-wide sell-through rate, this represents approximately 17.4 days of supply — a reasonable buffer for a retail network of this size, but one that masks significant SKU-level imbalance.

The fastest-moving SKUs carry as little as 6.7 days of supply — a stockout risk if replenishment lead times exceed roughly a week — while the slowest-moving SKUs sit at nearly 58 days of supply, tying up working capital in inventory that is turning over roughly 8 times slower than the fastest SKUs. Neither extreme is efficient: the low-end risks lost sales, and the high-end risks markdowns, shrinkage, or simply capital that could be redeployed elsewhere.

- SKU-level days-of-supply should become a standing Power BI measure, with conditional formatting to flag both under-7-day (stockout risk) and over-45-day (overstock) SKUs.
- The \$300K currently tied up in inventory is a meaningful working-capital figure for a network this size — a modest rebalancing toward faster-moving SKUs could free up capital without adding stockout risk.
- Because Toys is both the fastest-turning category on average and the most stockout-prone, it is the strongest candidate for increased reorder frequency rather than increased order size.

Key Findings

- Electronics is the hidden margin leader — 44.6% margin from just 3 SKUs, nearly double the margin of Toys, the network's highest-revenue category.
- Downtown stores outperform every other location type in every single product category — the strongest, most consistent driver of profit variance in the dataset.
- December is the single most important month of the year, and the July–August trough is entirely predictable — both years show the identical pattern.
- Weekends (Fri–Sat) generate 39% of all revenue from just 2 of 7 days — a strong signal for staffing and replenishment scheduling.
- Toys carries the highest stockout rate (7.4%) of any category while also being the highest-revenue category — the single biggest identified source of lost sales in the network.
- Inventory is unevenly distributed: some SKUs carry under 7 days of supply while others carry nearly 58 — a purchasing-cadence problem, not a total-inventory problem.

Strategic Recommendations

1. Expand the Electronics assortment

With a 44.6% margin on only 3 SKUs, Electronics is under-represented relative to its profitability. Testing 2–3 additional Electronics SKUs in top-performing Downtown stores is a low-risk way to grow high-margin revenue.

2. Prioritize Downtown-format stores in expansion plans

Downtown's category-agnostic outperformance suggests the growth lever is store format and footprint, not merchandising mix. Future store openings and renovation capital should weight Downtown locations first.

3. Build a stockout-alerting view in Power BI

A category- and store-filterable stockout KPI, with Toys and Art & Crafts flagged by default, would let regional managers close the highest-value gaps first — directly addressing the category most exposed to both stockout frequency and revenue loss.

4. Rebalance replenishment cadence by SKU velocity

Rather than adjusting total inventory volume, shift toward more frequent, smaller replenishment orders for fast-moving SKUs (under 10 days of supply) and reduced order frequency for slow movers (over 45 days) — freeing working capital without adding stockout risk.

5. Lock in December and pre-empt the summer trough

Given the identical seasonal pattern across both years in the dataset, promotional and staffing calendars can be planned with high confidence: maximize December readiness and use the July–August trough as a scheduled clearance or inventory-rebalancing window rather than reacting to it as a surprise dip.

Appendix: Category Performance Summary

Full-period revenue, profit, and margin by product category.

Category	Revenue	Profit	Margin
Toys	\$5,093,241	\$1,079,527	21.2%
Art & Crafts	\$2,705,364	\$753,354	27.8%
Games	\$2,226,836	\$673,993	30.3%
Sports & Outdoors	\$2,172,360	\$505,718	23.3%
Electronics	\$2,246,771	\$1,001,437	44.6%

Data Sources: sales.csv (829,262 records), products.csv (35 SKUs), stores.csv (50 locations), inventory.csv (1,593 stock records), calendar.csv — Maven Toys dataset, Maven Analytics (public domain). Period: January 1, 2022 – September 30, 2023.